

MENGYU HAN

Full Stack and Cloud Software Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Full-stack software engineer with **5+ years at SLB**, delivering cloud applications with Angular, C#/Go and Kubernetes. Solves product-critical algorithm and visualization problems, from patented carbon-storage site selection to interactive, **multi-million-cell reservoir rendering**.

PROFESSIONAL EXPERIENCE

SLB (Schlumberger) | Beijing, China

2021 - Present

Senior Frontend Engineer 2025 - Present | **Frontend Engineer** 2023 - 2025

Full Stack Developer 2021 - 2023

Prism | Geological Interpretation and Reservoir Visualization

Tech stack: Angular, TypeScript, RxJS, Three.js, WebGL/GLSL, Canvas, Webpack Module Federation; C#/.NET REST APIs; AKS, OpenTelemetry, Prometheus, Grafana

- **Dip Picking:** Led a 3-person team to deliver interactive picking and editing, combining coordinate transforms, sinusoid regression and true dip/azimuth calculations. Resolved a longstanding interpretation bottleneck; **adopted by 3 products**.
- **Reservoir rendering:** Replaced geometry-heavy rendering with GPU volume ray casting to enable interaction with multi-million-cell grids. Improved frame rates from **0-15 FPS to consistent 60+ FPS** on the same data and hardware.
- **Data loading:** Built backend chunking, streaming and encrypted OPFS caching for **~9M-element datasets**. Cut loading from **~3-5 min to ~30-40 s cold / ~5 s cached**, depending on network conditions.
- **Platform development:** Developed Module Federation microfrontend plugins and C# REST APIs. Contributed frontend/backend observability using OpenTelemetry, Prometheus and Grafana for workloads on AKS.
- **Technical delivery:** Resolved **13+ vendor integration issues**, guided contractor implementation and trained **12 engineers** across two cross-team sessions.

Related public PoCs: [Reservoir Volume Lab](#) | [Raymarching demo](#) | [Hex-grid demo \(code\)](#)

CarbonSCORE 2.0 | Carbon Storage Site Selection

Tech stack: Angular, TypeScript, RxJS, AG Grid, C#/.NET, REST APIs, AHP, TOPSIS

- **Algorithm and invention:** Co-developed AHP + TOPSIS site-selection logic with domain experts, resolving a product-delivery blocker and reducing user comparisons from **$O(n^2)$ to $O(n)$** . The algorithm was adopted in CarbonSCORE 2.0; named co-inventor on [U.S. Patent 12,404,748](#).
- **Full-stack delivery:** Implemented C# RESTful APIs for site scoring, criteria weights and inconsistency checks, plus the Angular/RxJS frontend and customized AG Grid interface. Delivered the prototype that guided product development.

Cybertron | Cloud Application Development

Tech stack: Angular, TypeScript, NgRx, RxJS, C#/.NET, Go (Golang), REST/OpenAPI, Docker, Kubernetes, GCP, Taiko + Gauge

- **Frontend:** Delivered Angular interfaces and reusable components for data management and visualization workflows. Used NgRx for shared application state and RxJS for asynchronous data coordination.
- **Backend:** Built C# RESTful APIs for Techlog-to-OSDU data ingestion and asynchronous job tracking; implemented Go services with concurrent task handling and OpenAPI documentation.
- **Cloud and testing:** Supported deployment and troubleshooting of C#/Go workloads in GCP Kubernetes clusters. Developed Taiko + Gauge UI automation and API/service tests to validate application workflows.

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ADDITIONAL EXPERIENCE

Enterprise Application Modernization | SLB

Tech stack: Angular, SQL Server, Oracle, IIS, Windows Server, DNS, HTTPS/TLS

- Upgraded a legacy application from **Angular 7 to 16** and coordinated a **Windows Server/IIS migration** with contractors, resolving compatibility and deployment issues.
- Maintained SQL Server/Oracle databases, HTTPS certificates and production configuration; extracted application-usage data for management reporting.

AWARDS AND RECOGNITION

- **Out of Ordinary (O2) - 2 awards** | Q4 2024 and Q2 2026.
- **BGC Innovation Carnival - Best Presentation, 2024** | Hand-gesture 3D visualization; 102 submissions.
- **Euro Hackathon - 1st Place, 2024** | Gesture-controlled log viewer; 54 submissions.
- **BGC Coding League - 3rd Place, 2025** | Frontend/game development; 24 teams.
- **Collaboration in Innovation, Q1 2026** | Cross-center 3D Gaussian Splatting integration with Spark.js and Three.js.
- **Other recognition:** IE Product Line Quarterly Star (Q2 2021); BGC Bootcamp Best Classroom Performance (2022).

VOLUNTEER AND COMPANY EVENT PROJECTS

2024 IDPwD | Sign Language Game | [Demo](#) | [GitHub](#)

Angular, MediaPipe. Designed and built a webcam-based sign-language game for SLB's disability initiative, helping participants explore sign language and communication accessibility.

WhatIf Studio | SLB100 Family Day | [UI preview](#) | [GitHub](#)

Next.js, Python, InsightFace/inswapper. Built a face-swap experience for SLB's centennial event, integrating a portrait gallery, face uploads and AI generation. Public preview shows the interface.

SLB100 Family Day | Gesture-Controlled 3D Game | [Demo](#) | [GitHub](#)

React, Three.js, MediaPipe. Developed a camera-driven 3D game with gesture-sequence recognition, hold-duration checks and animated challenges for company anniversary participants.

PATENTS AND PUBLICATIONS

U.S. Patent 12,404,748 | Co-inventor | Granted Sep 2025
[Carbon Sequestration and Storage Site Selection and Usage](#)

IEEE ICSSSM 2019 | Xiukun Zhao, **Mengyu Han**, Qianzhou Deng, Kelei Xue
[Decisions on supply and demand of bicycle-sharing service based on cobweb model](#)

EDUCATION

Washington University in St. Louis | Master of Information Systems Management, 2019
Tianjin University of Finance and Economics | B.S. Management Science, 2018